

**AMENDMENT NO. 1 FEBRUARY 2010
TO
IS 15684 : 2006 EXTENDERS FOR USE IN SYNTHETIC
RESIN ADHESIVES (UREA-FORMALDEHYDE) FOR
PLYWOOD — CODE OF PRACTICE**

(Page 1, clause 2) — Delete the clause and renumber subsequent clauses.

(Page 1, clause 5, fifth line) — Substitute ‘Annex A’ for ‘IS 1508’.

(Page 2, clause 6) — Insert the following Annex after the clause:

**‘ANNEX A
(Clause 5)**

A-1 DETERMINATION OF FINENESS OF GRANULES

A-1.1 Procedure

A-1.1.1 Pass 150 g of the sample, dried to a constant weight in an air oven maintained at 100 to 105°C, successively through IS Sieve 100 and IS Sieve 80.

A-1.1.2 Weigh each powder, dried to a constant weight, separately after passing through the respective sieves.

A-1.2 Calculation

A-1.2.1 Calculate the percentage of the sample which has passed through each sieve, using the following formula:

$$\text{Percentage of sample passing through a particular sieve} = \frac{100 b}{a}$$

where

b = weight in g of that part of the oven dry sample which passes through the particular sieve, and

a = weight in g of the oven dry sample.

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A-2 DETERMINATION OF MOISTURE CONTENT

A-2.1 Procedure

A-2.1.1 Weigh accurately 3 g of the sample in a tarred, flat weighing bottle of glass, aluminium, nickel or stainless steel. Dry the material at a temperature of 100° to 105°C and cool in a desiccator and determine the correct weight of the material. Repeat drying, cooling and weighing till the weight is constant.

A-2.2 Calculation

A-2.2.1 Calculate as follows:

$$\text{Moisture, percent by weight} = \frac{\text{Loss in weight}}{\text{Oven – dry weight of sample}} \times 100$$

A-3 DETERMINATION OF ASH CONTENT

A-3.1 Procedure

A-3.1.1 Take about 5 g of the sample in a weighed porcelain or silica crucible. Place it in an air-oven maintained at 100° to 105°C, till a constant weight is attained. Then put the crucible on a clay-pipe triangle over a tripod stand and incinerate the substance at a low heat with a low Bunsen flame. After the initial decomposition is over, gradually raise the flame to its full capacity and heat till a carbon-free ash is left in the crucible. Finally, heat the crucible along with the ash for 20 min over a blowpipe or in a muffle furnace maintained at 750°C cool in a desiccator, and weigh.

A-3.2 Calculation

A-3.2.1 Calculate the percentage of ash by using the following formula:

$$\text{Ash, percent by weight} = \frac{100 b}{a}$$

where

b = weight in g of the ash, and

a = weight in g of the oven dry sample taken.

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A-4 DETERMINATION OF FAT CONTENT

A-4.1 Apparatus

A-4.1.1 The apparatus consists of a fat-extraction tube made of heat-resisting glass, conforming to the details given in Fig. 1 fitted either with a two-hole bark cork as shown in Fig. 1A or a siphon as shown in Fig. 1B and provided also with a solid bark cork or ground glass stopper. The top of the neck is flared as shown in Fig. 1A and Fig. 1B to form a channel between the cork or stopper and the top of the neck. The narrow tube terminating in a hook shape at its lower end is a sliding-fit in the cork and long enough for opening at its lower end to be situated, if necessary, at a distance of 25 mm from the bottom of the tube in which it is enclosed.

NOTE — The length of the tube from the base to the bottom of the shoulder given as 200 ± 5 mm in Fig.1 may be varied, if desired, for example, to obtain a tube of less overall length to fit a centrifugal machine. If this is done, the diameter of the tube should be such that the volume of the liquid required to fill the tube to the base of the shoulder is still 105 ± 5 ml.

A-4.2 Reagents

A-4.2.1 *Quality of Reagents* — Unless specified otherwise, pure chemicals shall be employed in tests and distilled water shall be used (*see* IS 1070) where the use of water as a reagent is intended.

NOTE — 'Pure chemicals' shall mean chemicals that do not contain impurities which affect the results of the analysis.

A-4.2.2 The following reagents are required:

- a) *Hydrochloric Acid* — density 1.122 g/ml at 20°C (sp gr 1.125, conforming to IS 265).
- b) *Ether* — density 0.718 g/ml at 20°C (sp gr 0.720, conforming to IS 336).
- c) *Light Petroleum* — boiling range 40 to 60°C.
- d) *Ethyl Alcohol* — 96 percent (conforming to IS 323).

A-4.2.3 The alcohol, ether and light petroleum shall leave no residue when 100 ml are evaporated and dried at 101° to 102°C.

A-4.3 Procedure

A-4.3.1 Place 15 ml of hydrochloric acid in the fat-extraction tube. Weigh

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accurately about 5 g of the sample and carefully transfer to the extraction tube. Shake the tube so as to wet and wash down any particles that might have adhered to the sides. Cautiously heat the contents of the tube over a Bunsen flame, taking care to avoid frothing and making sure that all particles are dissolved in the acid. Finally boil for 10 min.

All dimensions in millimetres.

FIG. 1 FAT-EXTRACTION TUBES

A-4.3.2 Cool the tube and contents in running water until a temperature of $27 \pm 2^{\circ}\text{C}$ is reached. Add 10 ml of alcohol and mix thoroughly. Add 25 ml of ether, close the tube with water-wetted cork (extracted with ether before use) or stopper of good quality and shake vigorously for 30 s. Remove the cork and wash the cork and neck of the tube with 25 ml of light petroleum, so that the washings run back into the tube. Replace the cork, again water-wetted, and shake for 30 s. Complete extraction of the fat is dependent upon the satisfactory mixing at each stage. Allow to stand until the upper layer is clear and completely separated from the aqueous layer (usually not less than 30 min), or whirl in a centrifugal machine at 1 000 rev/mm for 30 s. Remove the cork, wash the cork and the neck of the tube with ether, return the washings to the tube and

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blow off the ethereal layer into a clean, dry, 200-ml wide-mouthed flask, removing as much ether as possible. Wash the inside of the tube twice with 10 ml of ether, adding these washings to the ether previously blown off. Make three further extractions. It is essential that the cork (or stopper) be wetted with water before each insertion and washed with ether during each removal. Rubber stoppers shall not be used.

A-4.3.3 Cautiously distil off ether as it is collected in portions during the extraction process, clean the outside of the flask, then dry in an oven at 98° to 100°C to constant weight (approximately one hour), blowing out the ether vapour from the flask from time to time by means of a current of dust-free air. Add 20 ml of light petroleum. Allow to stand for 5 to 10 min, then pour off the light petroleum so as to retain any insoluble sediment in the flask and remove all traces of fat from the inside and the neck of the flask by means of a stream of light petroleum directed from a wash-bottle.

A-4.3.4 Carefully clean the outside of the flask and dry with precautions to remove vapour of the solvent as before, to a constant weight at 98° to 100°C.

A-4.4 Calculation

A-4.4.1 Calculate as follows:

$$\text{Fat, percent by weight} = \frac{100 (W_2 - W_3)}{W_1}$$

where

W_2 = weight in g of the flask and fat,

W_3 = weight in g of the flask after light petroleum extraction, and

W_1 = weight in g of the oven dry sample taken.

A-5 DETERMINATION OF NITROGEN CONTENT

A-5.1 Reagents

A-5.1.1 *Quantity of Reagents* — Unless specified otherwise, pure chemicals shall be employed in tests and distilled water (*see* IS 1070) shall be used where the use of water as a reagent is intended.

NOTE — 'Pure chemicals' shall mean chemicals that do not contain impurities which affect the results of analysis.

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A-5.1.2 The following reagents are required:

- a) *Potassium Sulphate* — crystallized, nitrogen-free powdered.
- b) *Sodium Hydroxide Solution* — 50 percent by weight, prepared from sodium hydroxide conforming to IS 376.
- c) *Standard Sodium Hydroxide Solution* — 0.2 N.
- d) *Concentrated Sulphuric Acid* — concentrated, nitrogen-free (sp gr 1.84, conforming to IS 266).
- e) *Copper Sulphate*
- f) *Standard Sulphuric Acid* — 0.2 N.
- g) *Methyl Red Indicator* — 0.02 percent solution in neutral alcohol.

A-5.2 Procedure

A-5.2.1 Weigh accurately 1.0 g of the sample and approximately 10 g of potassium sulphate and transfer completely to a 300-ml Kjeldahl flask. Add 20 ml of concentrated sulphuric acid and 0.2 g of copper sulphate. Gently rotate the flask so that all the solids are well wetted.

A-5.2.2 Place the flask on metal frame so that the neck is inclined at 45° to the horizontal and the bulb rests on an asbestos sheet with a hole of such size that the flame does not touch the flask above the level of the liquid, and heat to gentle boiling for 10 min. At the end of this time, boil the contents of the flask briskly until the yellowish colour has disappeared and subsequently for 15 min.

A-5.2.3 Allow the liquid to cool, dilute with approximately 200 ml of neutral distilled water or deionized water conforming to IS 1070, transfer to a 1 000 ml round-bottomed flask and add a few pieces of granulated zinc or a small amount of powdered pumice to prevent bumping. Fit the flask with splash head and tap funnel. Connect it to a condenser, pipette out 40 ml of sulphuric acid into a 600-ml conical flask and place under the outlet of the condenser so that the outlet of the condenser is below the level of the acid. An absorption bulb trap shall be provided on the conical flask to avoid flask splashings.

A-5.2.4 Run into the round-bottomed flask 100 ml of sodium hydroxide solution through the tap funnel. Heat by means of a steady flame so that boiling is maintained at a uniform rate. Collect about 100 ml of the distillate in the conical flask. After 30 min, lower the conical flask so that the end of the condenser is above the level of the liquid. Continue distillation for another 3 min and then remove the flame from under the round-bottomed flask. Allow the condenser to drain for 2 to 3 min and finally wash it with distilled water or deionized water allowing the wash water to drain out of the condenser.

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A-5.2.5 Titrate the excess acid with 0.2 N sodium hydroxide solution, using methyl red as indicator.

A-5.2.6 Carry out a blank determination by taking 1.0 g of sucrose in place of the sample under test, using the same quantities of reagents and the same conditions of test.

A-5.2.7 Deduct the quantity of acid required in the original test from that required in the blank test (one ml of 0.2 N standard sulphuric acid = 0.002 8 g of nitrogen).

A-5.3 Calculation

A-5.3.1 Calculate the percentage of nitrogen on moisture, fat and ash-free basis by the following formula:

$$\text{Nitrogen, percent by weight} = \frac{10\,000\,N}{W(100 - M - A - F)}$$

where

N = calculated weight of nitrogen (*see A-5.2.7*);

W = weight in g of the sample taken for the test (*see A-5.2.1*);

M = moisture, percent by weight;

A = ash, percent by weight; and

F = fat, percent by weight.

A-6 DETERMINATION OF ACIDITY

A-6.1 Procedure

A-6.1.1 Weigh accurately about one gram of the sample into a 250-ml graduated flask and run in 50 ml of 0.1 N standard sodium hydroxide solution from a pipette, constantly agitating the flask. Stopper the flask and continue the agitation for 15 minutes. Make up the volume to 250 ml with distilled water or de-ionized water (neutral to phenolphthalein) conforming to IS 1070. Shake the flask vigorously and allow any solid matter to settle. From the clear supernatant liquid, pipette out 50 ml and titrate with 0.05 N acid, using alcoholic phenolphthalein (one gram per 100 ml) as indicator. The acid shall be run in with vigorous shaking to prevent any local precipitation of curd in the case of proteinaceous samples. The number of ml 0.1 N alkali used up by one gram of moisture, fat- and ash-free materials is termed the acidity of the sample.

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A-6.2 Calculation

A-6.2.1 Calculate the acidity of the material on moisture, fat- and ash-free basis by the following formula :

$$\text{Titration acidity in ml of 0.1 N sodium hydroxide solution per 100 g of the sample} = \frac{10\,000 (500 N_1 - 50 V N_2)}{W (100 - M - A - F)}$$

where

- N_1 = normality of standard sodium hydroxide solution;
- V = volume in ml of standard sulphuric acid required to neutralize 50 ml of the clear supernatant liquid;
- N_2 = normality of standard sulphuric acid;
- W = weight in g of the sample taken for the test;
- M = moisture, percent by weight;
- A = ash, percent by weight; and
- F = fat, percent by weight.

(CED 20)

Amendment No 1 to IS 5887 (Part 8/Sec 1) : 2002/ISO 6888-1 : 1999 (FAD)

Amendment No 1 to IS 5887 (Part 8/Sec 2) : 2002/ISO 6888-2 : 1999 (FAD)

Amendment No 1&2 to IS 6566 : 2001/ISO 668 : 1995 (TED)

Amendment No 3, 4 &5 to IS 13288 (Part 1) : 1993/ISO 1496-1 : 1990 (TED)

Amendment No 3&4 to IS 7622 : 2001/ISO 3874 : 1997 (TED)

**All 5 amendments mentioned above have been printed from ISO
amendments through manual pasting and hence no soft copy like others.**